

SquadHub Mobile Application

Usage Guideline & Technical Documentation

1. Introduction

SquadHub is an Android mobile application built using Flutter that enables users to collaborate within groups (“squads”). The application implements authentication, role-based access, group interaction, and content navigation using a modern Flutter architecture and Firebase-backed services.

This document serves as a practical user guide, detailing how to operate every implemented feature of the application, and briefly describes the tools and frameworks used.

2. Tools, Frameworks, and Technologies Used

2.1 Frontend

Flutter SDK (v3.38.x)

Cross-platform UI framework used to build the Android application.

Dart

Programming language used for application logic and UI composition.

2.2 Backend & Services

Firebase Authentication

Used for secure user login and session management.

Firebase Cloud Firestore

Stores user profiles, group data, and related application content.

Firebase Core

Provides backend initialization and configuration support.

2.3 Android Platform

Gradle (Android build system)

Manages Android builds and dependency resolution.

Android SDK

Used for packaging and deploying the APK.

3. User Credentials (Demo / Testing)

Group Admin Account

Email: `yonatan.dev.prime@gmail.com`

Password: `Yonatan@123`

Role: Group Admin

Standard User Account

Email: `amhakifle09@gmail.com`

Password: `Amha@123`

Role: User

4. Application Usage Guide (Step-by-Step)

4.1 Installing and Launching the App

1. Install the APK on an Android device.
2. Open the application from the device app drawer.
3. The Login Screen is displayed by default.

4.2 User Authentication

4.2.1 Logging In

1. On the Login Screen, enter a valid email address.
2. Enter the corresponding password.
3. Tap the Login button.

Expected Result:

- User is authenticated.
- User is redirected to the Home/Dashboard screen based on their role.

4.2.2 Handling Invalid Login

1. Enter an incorrect email or password.
2. Tap Login.

Expected Result:

- An error message is displayed.
- User remains on the login screen.

4.3 Home / Dashboard Navigation

After successful login:

The app displays the main dashboard.

Navigation is handled through visible UI components (buttons, cards, or navigation bars).

Available actions depend on user role.

4.4 Group Admin Functionalities

4.4.1 Viewing Group Information

1. Log in using the Group Admin credentials.
2. Navigate to the Group/Squad section.
3. Select a group to view details.

Expected Result:

Group name and related data are displayed.

4.4.2 Managing Group Content

1. Open the desired group.
2. Use available action buttons (add/view/manage).
3. Submit or update information as prompted.

Expected Result:

Group data updates are saved and reflected immediately.

4.5 Standard User Functionalities

4.5.1 Viewing Assigned Groups

1. Log in using a User account.
2. Navigate to the group listing section.
3. Select a group.

Expected Result:

User can view group content but cannot perform admin-level actions.

4.5.2 Interacting with Group Content

1. Open a group.
2. Browse available information or updates.

Expected Result:

User can read content but cannot modify restricted fields.

4.6 Logging Out

1. Open the profile or menu section.

2. Tap Logout

Expected Result:

User session is terminated.

App returns to the Login Screen.

5. Session and Data Handling

User sessions are securely managed using Firebase Authentication.

Data displayed in the app is fetched in real-time from Firestore.

Role-based UI restrictions are enforced at the application logic level.

6. Limitations (Based on Current Implementation)

Android platform only (no iOS or web builds included).

No offline data persistence implemented.

No password recovery feature currently available.

Roles are predefined and not modifiable within the app UI.

7. Conclusion

This documentation provides a complete, implementation-focused usage guide for the SquadHub application. All steps, credentials, and features described reflect the current working state of the app, making this document suitable for academic submission, demonstrations, and user onboarding.